12.4 Volumes of Prisms and Cylinders

Name _____ Period ____ Date _____

Geometry 12.4 · Student sheet

I can…

- find the volume of a prism or a cylinder.
- find a missing dimension from a volume.
- use the scale factor of similar solids to compare their volumes.

Vocabulary · fill in the missing word

Volume — the number of cubic units _____ in the interior of a solid

Cavalieri's Principle — if two solids have the same height and the same _____ area at every level, then they have the same volume

Similar solids — two solids of the same type with equal ratios of _____ linear measures

Part 1 · Discovery: lean the stack

Build a straight stack. Measure one card and the height of the stack. Then lean the stack.

	Area of one card	Height of the stack	Area × height
Straight stack			
Leaning stack			

Leaning the stack does not change its _____.

Part 2 · Notes and examples (fill in each note, then work the example)

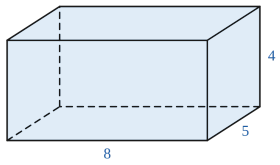
Volume of a prism: V = _____, where B is the area of a base and h is the height.

Example 1

Find the volume of each prism.

a. a rectangular prism 8 by 5 by 4 (shown)

b. a triangular prism with a right-triangle base with legs 6 and 8, and a length of 12



Example 2

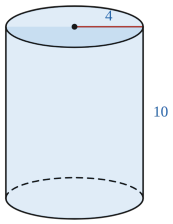
A rectangular prism has a volume of 360 cubic inches. Its base is 9 inches by 8 inches. Find its height.

Volume of a cylinder: V = _____

Cavalieri's Principle: an oblique prism or cylinder has the _____ volume as a right one with the same base and height.

Example 3

Find the volume of the cylinder. Would an oblique cylinder with the same radius and height have a different volume? Round to the nearest hundredth.



Example 4

A cylinder has a volume of 1000 cubic centimeters and a height of 10 centimeters. Find its radius. Round to the nearest hundredth.

Similar solids: if the scale factor is a : b, the volumes are in the ratio _____.

Example 5

Two cylinders are similar with a scale factor of 2 : 3. The smaller one has a volume of 80 cubic inches. Find the volume of the larger one.

Part 3 · Partner practice

Partner 1

Find the volume of a cube with edges of 7 centimeters.

Partner 2

A cylinder has a diameter of 6 and a height of 9. Find its volume. Round to the nearest hundredth.